KNECK OFFICIAL

Biology — Paper 2

231/2 • ORIGINAL REVISION SET 01 • 2 hours • 80 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all questions in Section A. In Section B answer question 6 and either question 7 or 8. Draw large, labelled diagrams where asked.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

Five questions, 8 marks each.

1. A potato cylinder initially weighs 5.0 g. After 1 hour: distilled water 5.6 g; dilute sucrose 5.0 g; concentrated sucrose 4.3 g. (a) Calculate percentage mass change in water and concentrated sucrose. (4) (b) Explain the changes using water potential. (4) [8 marks]

2. A pond food chain is algae → mosquito larvae → small fish → kingfisher. (a) Identify producer and secondary consumer. (2) (b) Explain why energy falls at each level. (3) (c) Predict two effects of a pesticide that kills larvae, and one limitation of this prediction. (3) [8 marks]

3. A couple are both carriers of an autosomal recessive condition (Aa). (a) Construct a genetic cross. (4) (b) Give probability of an affected child and probability of a carrier. (2) (c) Explain why each pregnancy has the same probability. (2) [8 marks]

4. Describe the route of an impulse in a spinal withdrawal reflex, and explain its protective value. [8 marks]

5. An enzyme reaction runs at 5,15,25,35,45,55°C with rates 1,3,6,10,4,0 arbitrary units. (a) Plot rate against temperature. (3) (b) State the highest measured rate temperature. (1) (c) Explain the rise and fall. (4) [8 marks]

SECTION B

Answer question 6 and either question 7 or 8.

6. A seedling experiment has mean heights (cm): day 0,2,4,6,8,10; batch A in light: 1,2,4,7,11,15; batch B in dark: 1,3,7,12,15,16. (a) Draw both curves on the same axes. (6) (b) Calculate each batch mean growth rate from day 2 to day 8. (4) (c) Explain why the dark-grown batch becomes tall and pale. (4) (d)

Suggest three controlled variables and a method of improving reliability. (4) (e) Why do these height data alone not prove greater biomass in batch B? (2) [20 marks]

7. Describe adaptations of a mammalian heart and circulatory system for effective transport. Include a labelled longitudinal heart diagram. [20 marks]

8. Explain sexual reproduction in a flowering plant from pollination to dispersal, and explain the importance of seed dormancy. [20 marks]

END OF QUESTIONS

MARKING GUIDANCE

Biology • Paper 2 • Set 01

Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (8 marks) Water +12%; concentrated sucrose -14%, each method/answer 2 (4). Water enters cells by osmosis from higher to lower water potential; concentrated solution draws water out across selectively permeable membranes (4).

2. (8 marks) Algae; small fish (2). Respiration/heat, uneaten parts, egestion/excretion (3). Larvae fall, fish food falls/possible algal increase; alternative diets or other interactions limit simple predictions (3).

3. (8 marks) Parental genotypes/gametes and Punnett square AA,Aa,Aa,aa (4). Affected 1/4; carrier 1/2 (2). Independent fertilisation events; previous children do not alter parental allele segregation (2).

4. (8 marks) Receptor detects stimulus (1), sensory neurone (1), dorsal root/CNS (1), relay neurone and synapse (1), motor neurone (1), effector muscle contracts (1), rapid involuntary response (1), limits tissue injury (1).

5. (8 marks) Axes/scale/points with curve (3). 35°C among tested values; exact optimum needs finer sampling (1). Rising kinetic energy and effective collisions (2); high temperature changes active-site shape/denaturation (2).

SECTION B

6. (20 marks) Axes with units 2, scale 1, correct points 2, key/smooth curves 1 (6). A=(11-2)/6=1.5 cm/day; B=(15-3)/6=2.0 cm/day (4). Etiolation/elongation using stored reserves, little chlorophyll synthesis in dark (4). Same seed species/age, water, nutrients, temperature; replicate and calculate mean (4). Elongation may reflect water/cell expansion, not net dry matter; weigh dry biomass (2).

7. (20 marks) Diagram labels chambers, major vessels and septum, directions correct (6). Four chambers prevent mixing (2); thick left ventricle for systemic pressure (2); valves prevent backflow (2); coronary vessels supply myocardium (2); double circulation gives efficient lung/systemic flow (2); capillary thin walls and network for exchange (2); elastic arteries/veins valves appropriate functions (2).

8. (20 marks) Pollination definition and agents (2); pollen germination/tube growth (3); male nuclei and double fertilisation producing zygote/endosperm (4); ovule→seed, ovary→fruit (2); embryo/seed coat formation (2); dispersal mechanisms with adaptations, three examples (6); dormancy enables survival until favourable germination conditions (1).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf